

Q1 - 24 January - Shift 1

Given below are two statements :

Statements I : The temperature of a gas is -73°C .

When the gas is heated to 527°C , the root mean square speed of the molecules is doubled.

Statement II : The product of pressure and volume of an ideal gas will be equal to translational kinetic energy of the molecules.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both statement I and Statement II are true
- (2) Statement I is true but Statement II is false
- (3) Both Statement I and Statement II are false
- (4) Statement I is false but Statement II is true

Space for your notes:

Q2 - 25 January - Shift 1

The root mean square velocity of molecules of gas is

- (1) Proportional to square of temperature (T^2).
- (2) Inversely proportional to square root of temperature $\sqrt{\frac{1}{T}}$.
- (3) Proportional to square root of temperature $\sqrt{T}$.
- (4) Proportional to temperature (T).

Space for your notes:

Q3 - 29 January - Shift 2

At 300 K, the rms speed of oxygen molecules is

$\sqrt{\frac{\alpha + 5}{\alpha}}$ times to that of its average speed in the

gas. Then, the value of α will be (used $\pi = \frac{22}{7}$)

- (1) 32
- (2) 28
- (3) 24
- (4) 27

Space for your notes:

Q4 - 30 January - Shift 1

The pressure (P) and temperature (T) relationship of an ideal gas obeys the equation $PT^2 = \text{constant}$. The volume expansion coefficient of the gas will be:

- (1) $3T^2$ (2) $\frac{3}{T^2}$
(3) $\frac{3}{T^3}$ (4) $\frac{3}{T}$

Space for your notes:

Q5 - 30 January - Shift 2

A flask contains hydrogen and oxygen in the ratio of 2 : 1 by mass at temperature 27°C . The ratio of average kinetic energy per molecule of hydrogen and oxygen respectively is :

- (1) 2 : 1 (2) 1 : 1
(3) 1 : 4 (4) 4 : 1

Space for your notes:

Q6 - 01 February - Shift 1

The average kinetic energy of a molecule of the gas is

- (1) proportional to absolute temperature
(2) proportional to volume
(3) proportional to pressure
(4) dependent on the nature of the gas

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2)

Q2 (3)

Q3 (2)

Q4 (4)

Q5 (2)

Q6 (1)

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Q1 (2)

Statement-I

$$T_1 = -73^\circ\text{C} = 200 \text{ K}$$

$$T_2 = 527^\circ\text{C} = 800 \text{ K}$$

$$\frac{V_1}{V_2} = \frac{\sqrt{\frac{3RT_1}{M}}}{\sqrt{\frac{3RT_2}{M}}} = \sqrt{\frac{T_1}{T_2}}$$
$$= \sqrt{\frac{200}{800}} = \frac{1}{2}$$

$$V_2 = 2V_1 \text{ (True)}$$

Statement-II

$$PV = nRT$$

$$\text{Translational KE} = \frac{3}{2}nRT \text{ (False)}$$

Q2 (3)

The rms speed of a gas molecule is

$$V_{\text{RMS}} = \sqrt{\frac{3RT}{M}}$$

$$V_{\text{RMS}} \propto \sqrt{T}$$

Q3 (2)

$$\sqrt{\frac{3RT}{M}} = \sqrt{\frac{\alpha + 5}{\alpha}} \sqrt{\frac{8}{\pi} \frac{RT}{M}}$$

$$3 = \frac{\alpha + 5}{\alpha} \frac{8}{\pi}$$

$$\alpha = 28$$

Ans. 2

Q4 (4)

$PT^2 = \text{constant}$, Using $PV = nRT$

$$P = \frac{nRT}{V}$$

$$PT^2 = \frac{nRT}{V} \times T^2 = \text{constant}$$

$$\Rightarrow T^3 = KV$$

$$\text{So, } \frac{d}{dT}(KV) = 3T^2$$

$$\Rightarrow \frac{KdV}{dT} = 3T^2$$

$$\Rightarrow dV = \frac{3T^2}{K} dT$$

$$dV = V \gamma \frac{dT}{T}$$

$$\Rightarrow \gamma V = \frac{3T^2}{K} \Rightarrow \gamma = \frac{3T^2}{KV} = \frac{3T^2}{T^3} = \frac{3}{T}$$

Q5 (2)

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$$K_{av} = \frac{5}{2} kT$$

$$\text{Ratio} = 1 : 1$$

Q6 (1)

Basic theory

Translational *K.E* on average of a molecule is $\frac{3}{2}$ KT which is independent of nature, pressure and volume.

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